

Analysis First-Year Exam, January 2025, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let (a_n) be a bounded sequence of strictly positive real numbers. Prove that

$$\limsup(a_n^{1/n}) \leq \limsup(a_{n+1}/a_n).$$

Hence prove that if $a_{n+1}/a_n \rightarrow \ell \in \mathbb{R}$ then $a_n^{1/n} \rightarrow \ell$ (as $n \rightarrow \infty$).

2. The diameter D of the bounded metric space X is defined by

$$D = \sup\{d(a, b) : a, b \in X\}.$$

Prove that if X is compact then $D = d(a_0, b_0)$ for some $a_0, b_0 \in X$.

3. The compact metric space X contains $A_1 \supseteq A_2 \supseteq \dots$ as a decreasing sequence of closed sets.

(i) Prove that if each A_n is nonempty then so is the intersection $A = \bigcap_{n=1}^{\infty} A_n$.

(ii) Prove that if $U \subseteq X$ is open and contains A then U contains some A_n .

4. Let $f : X \rightarrow Y$ be a map between metric spaces and consider the statements. Does (a) imply (b)? Does (b) imply (a)? In each case, give a proof if so, a counterexample if no.

(a) if (x_n) is any Cauchy sequence in X then $(f(x_n))$ is Cauchy in Y ;

(b) f is uniformly continuous.

5. Let $\ell \in \mathbb{R}$, let $f : \mathbb{R} \rightarrow \mathbb{R}$, and assume that $f'(t) \rightarrow 0$ as $\mathbb{R} \ni t \rightarrow \infty$. Prove that if $f(n) \rightarrow \ell$ as $\mathbb{N} \ni n \rightarrow \infty$ then $f(t) \rightarrow \ell$ as $\mathbb{R} \ni t \rightarrow \infty$.